



Jaypee Institute of Information Technology, Noida

Department of Computer Science & Engineering and IT

B.TECH MAJOR PROJECT



An AI workspace for teachers to set papers and check answer sheets from their own course material

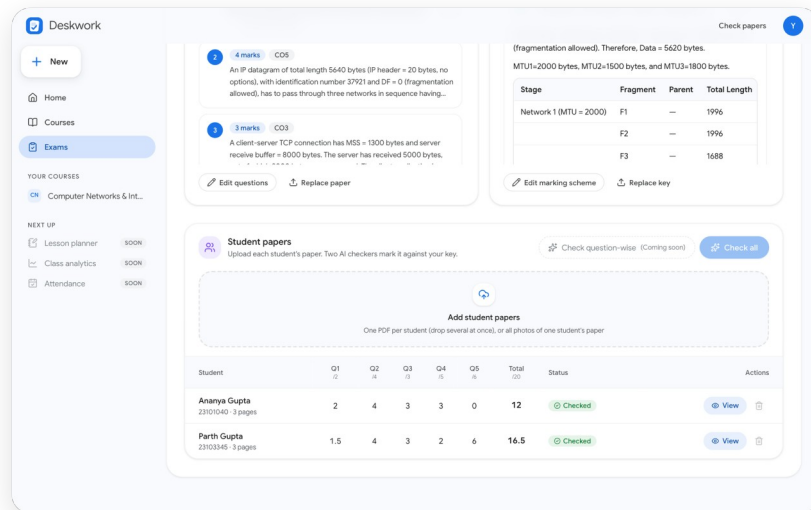
SUBMITTED BY

Yug Jindal
23103328

Anish Das
23103322

Yashita Gogia
23103305

Anjaneya Sharma
23103301



The problem, and what Deskwork does



Today

- Hours checking handwritten papers, one by one
- Assignments and papers written from scratch
- Tired eyes miss slips, marks drift
- Generic AI chat does not know your slides



With Deskwork

- Upload course slides once
- Generate assignments, solutions and papers with answer keys
- Two AI checkers mark each scanned paper against your key
- Disagreements flagged, teacher has the final say
- Marks to Excel in one click

Deskwork in action

Real CN & IoT slides and real handwritten answer sheets

Student papers

Every paper checked, marks per question

Add student papers

One PDF per student (drop several at once), or all photos of one student's paper

Student	Q1 (2)	Q2 (4)	Q3 (3)	Q4 (5)	Q5 (6)	Total (20)	Status	Actions
Ananya Gupta 23101040 - 3 pages	2	4	3	3	0	12	Checked	View
Parth Gupta 23103345 - 3 pages	1.5	4	3	2	6	16.5	Checked	View

Jaypee Institute of Information Technology, Noida

T2 Examination

Course Title: Computer Networks & Internet of Things

Course Code: 18B11CS311

Maximum Time: 1 Hour

Maximum Marks: 20

Note: Attempt all Questions.

Course Outcomes (CO) Table

Course Outcome	Description
CO-1	Identify and explain various application protocols.

Question paper written from the slides

Both checkers' marks, with the reason

WHAT THE STUDENT WROTE

Case 1: GBN total transmissions = 37 (incorrect step counts), SR total transmissions = 29.

WHY THESE MARKS

SR total is correct (29). GBN total is given as 37 by the student, whereas the key states 39. Let's verify student's GBN step 1 to 4: 6 lost → 7 discarded, after retransmit [6 7 8 9 10 11 12] → 12 lost → 7-11 discarded; retransmit [12 13 14 15 16 17 18] → 18 lost → 13-17 discarded; retransmit [18 19 20 21 22 23 24] → 24 lost → 19-23 discarded. Student total is 37 which is incorrect (key is 39). So GBN gets 0 marks, SR gets 1.0 mark.

MISTAKES FOUND

- Incorrect total transmissions and step-by-step count for GBN (wrote 37 instead of 39)

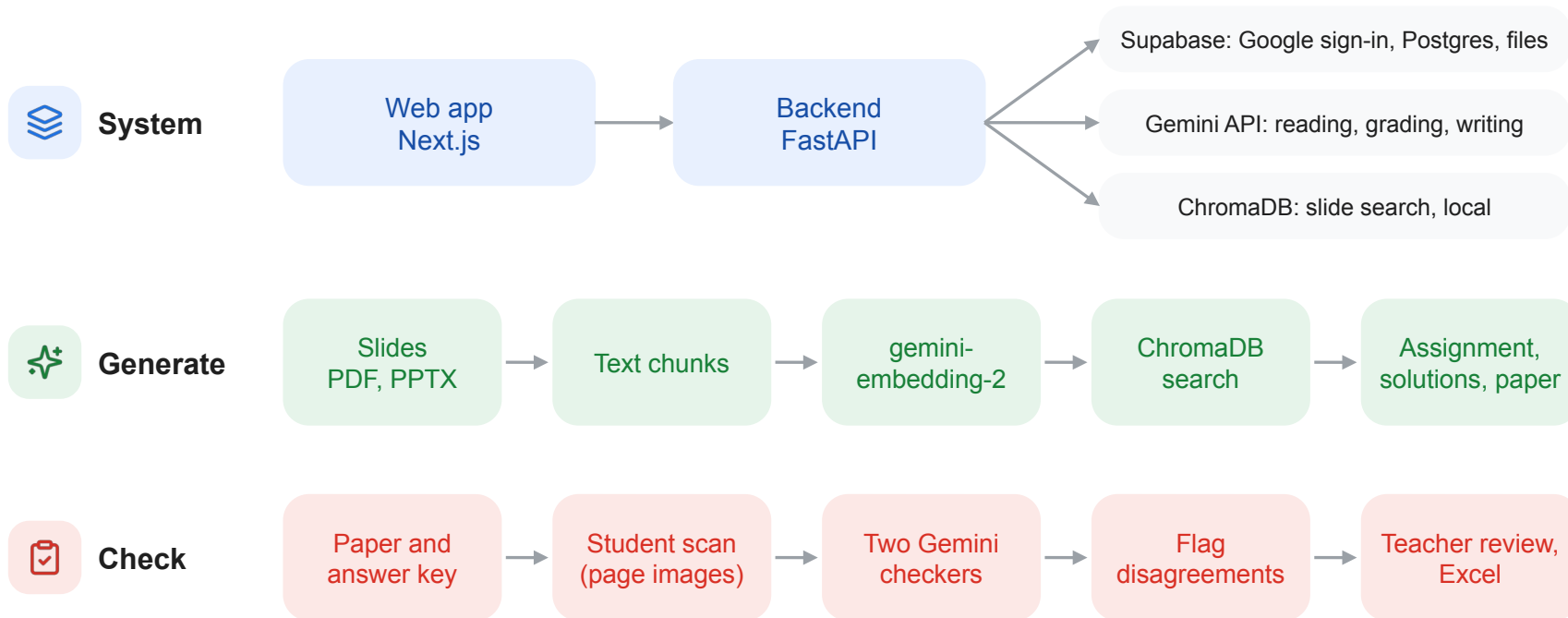
Imagine a user typing characters into a remote terminal session (like Telnet or SSH) over a slow connection:

- Without Nagle's Algorithm:** Every time the user presses a key, a new 1-byte TCP segment is created and sent immediately. If a user types the word "CAT" (3 characters), it generates 3 separate packets, resulting in heavy header overhead for very little data.
- With Nagle's Algorithm:**
 - The user types 'C'. Because no unacknowledged packets are in flight, 'C' is sent immediately in a TCP segment.
 - The user types 'A' and 'T' shortly after. Because the ACK for the packet containing 'C' has not arrived yet, the sender buffers 'A' and 'T'.
 - Once the ACK for 'C' comes back from the receiver, the sender packages the buffered bytes ('AT') into a single segment and transmits them together.

This significantly reduces the number of small packets on the network.

Answers cite the slide pages used

How it works



Tech stack



Next.js 16, React 19

Web app, Tailwind, Motion



FastAPI, Python 3.12

API, grading, PDF and PPTX reading



Supabase

Google sign-in, Postgres, file storage



Gemini 3.5 + 3.1 Flash-Lite

Reads scans, marks, writes papers



Embeddings + ChromaDB

gemini-embedding-2, slide search (RAG)



Pytest + Playwright

Unit, end-to-end and browser tests

Impact

2

AI checkers on every answer

1

slide upload, used by every chatbot

16.5/20

real booklet checked, slips caught

35/35

end-to-end checks on real material



Less time on repetitive checking and paper setting



Consistent marks, each with its reason shown



Everything built from the teacher's own slides

References



Design

- Material Design 3, m3.material.io
- Google Meet and Workspace UI (layout, pill navigation)
- Google Sans typeface, fonts.google.com
- Lucide icons, lucide.dev



Research

- arXiv 2604.22774: models silently fix student slips when transcribing
- arXiv 2603.13083: confidence alone is no safe flag
- PNAS Nexus 2026: teachers over-trust AI grades



Docs

- Gemini API: structured output, embeddings, ai.google.dev
- Supabase: Auth, Postgres, Storage, supabase.com/docs
- ChromaDB, docs.trychroma.com
- Next.js, nextjs.org/docs; FastAPI, fastapi.tiangolo.com



Products studied

- Gradescope (grading by question)
- MagicSchool (teacher content tools)



Future scope

What we will add by the final evaluation



Check question-wise

Mark one question for the whole class at once



Class analytics

Per-question scores and weak topics



Lesson planner

Lecture plans drafted from the course slides



Question bank

Reuse questions across papers and exams



Thank you